

Johannes Roth

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Machine Learning / Research Engineer and PhD researcher with 6+ years of experience building large-scale data pipelines, embedding systems, dataset curation workflows, and model evaluation tools. Strong background in data-centric ML and research engineering: processing hundreds of millions to billions of images, extracting and comparing model representations, designing active sampling methods, building validation/evaluation workflows, and releasing reproducible datasets and tooling.

EXPERIENCE

PhD Researcher - NeuroAI / Computational Cognitive Neuroscience *May 2022 – Present*

Max Planck Institute for Human Cognitive and Brain Sciences & University of Giessen

- Built large-scale ML pipelines to score and filter **2.1B images**, curating **LAION-Natural** and **ReLAION-2B Natural** with quality control, dataset documentation, and reproducible public release workflows.
- Generated and released **167GB** of ViT-H/14 embeddings to support visual similarity search, dataset construction, and model-comparison experiments over large unstructured image data.
- Developed **Natural Controversial Stimuli**, an active sampling framework that selects examples to maximize disagreement between model-based similarity judgments; relevant to data quality research, validation-set construction, and ablation-style analysis of data choices.
- Implemented embedding-based retrieval utilities over ViT/CLIP features to inspect nearest neighbours, compare model representations, and select hard examples from large image pools.
- Built and managed the **AWS-based data infrastructure** for **re:vision**, including dataset hosting, access workflows, and the public challenge website for reproducible model and dataset evaluation.

Research Assistant - ML in Medicine

Jun 2021 – May 2022

ScaDS.AI Dresden/Leipzig

- Built reproducible deep learning workflows for medical image segmentation and clinical prediction, covering pre-processing, GPU training, model comparison, and uncertainty-aware evaluation.
- Engineered a multi-plane **UNet++ ensemble** for glioblastoma segmentation (BraTS 2021), integrating Dice and Boundary loss to achieve competitive segmentation performance.
- Developed attention-based prediction models using **FT-Transformer** and **SAINT**, including epistemic uncertainty estimation for safer medical decision support.

Data Scientist / ML Engineer (Working Student)

Oct 2019 – May 2021

CHECK24 (Travel Vertical)

- Built and deployed a Flask/Redis-based image-processing service that applied ML predictions and metadata lookup across **>20 million** hotel images; optimized models via **ONNX** for production hosting.
- Improved recommendation and ranking workflows using Bayesian hyperparameter optimization, ranking metrics, and conversion-oriented evaluation.
- Built Grafana dashboards for API health, data quality monitoring, and production visibility in a cross-functional product engineering environment.

Data Scientist (Working Student)

Oct 2018 – Oct 2019

Webdata Solutions GmbH (now Vistex)

- Revamped a product-matching pipeline with neural networks trained on self-collected scraped data, increasing matching accuracy from **<50% to 92%**.
- Built data collection, training, evaluation, and interpretability workflows for product similarity, entity resolution, and noisy real-world catalog data.

Full-stack Developer (Freelance)

Oct 2020 – May 2021

Kimetric UG

- Implemented academic websites using **Django**, Nginx, and Unicorn; configured Linux hosting environments and automated deployment scripts.

PROJECTS & OPEN SOURCE

- **thingsvision** (Core Contributor) – Modular feature-extraction library for computer vision; used to extract and compare features from **100+** vision models including CLIP, ViT, and CNN families.
- **ReLAION-2B Natural** – Large-scale naturalistic image subset and embedding release designed for data selection, retrieval, and model comparison workflows.
- **LAION-fMRI / re:vision** – Contributed stimulus selection and release infrastructure for a deeply sampled **7T fMRI** dataset with **25,052** natural images and **165** fMRI sessions; supported AWS-backed access and public challenge tooling.

EDUCATION

PhD Candidate - Computational Cognitive Neuroscience **May 2022 – Present**
Max Planck Institute for Human Cognitive and Brain Sciences & University of Giessen
Dissertation on efficient experimental design, large-scale visual datasets, and model–brain alignment using modern computer vision models and high-resolution fMRI.

M. Sc. Computer Science **2017 – 2021**
Leipzig University
Grade **1.2**. Focus: Data Analysis, Machine Learning, Medical Image Processing.

- **Master's Thesis (Grade 1.1)** – Used GANs to synthesize stimuli that maximally activate targeted brain regions, re-covering known category-selective areas in human brains.

B. Sc. Business Information Systems **2014 – 2017**
Leipzig University
Grade **1.5**. Focus: Distributed Systems, E-Commerce, Data Management, Economics.

TECHNICAL SKILLS

Data-centric ML	Dataset curation, large-scale filtering, data quality, data selection, active sampling, validation data design, hard-example mining
Training & Evaluation	PyTorch, TensorFlow, scikit-learn, Transformers, automated model comparison, ranking metrics, uncertainty estimation, reproducible experiment workflows
Models	CLIP/OpenCLIP-style embeddings, ViT, CNNs, UNet++, GANs, foundation-model feature extraction and representation comparison
Engineering	Python, SQL, Docker, ONNX, AWS, Git, CI/CD, Flask, Redis, Linux, Bash, SLURM/HPC, Grafana
Languages	German (Native), English (Fluent)

SELECTED PUBLICATIONS & AWARDS

- **Award (2025)**: CMBB Replication Award for contribution to reliable coding practices in neuroscience.
- F.P. Mahner, **J. Roth**, et al. *Characterizing Universal Object Representations Across Vision Models*. arXiv preprint, 2026 (under review).
- **J. Roth**, M. N. Hebart. *How to sample the world for understanding the visual system*. CCN 2025.
- **J. Roth** et al. *Ten principles for reliable, efficient, and adaptable coding*. Communications Psychology 2025.

LEADERSHIP & COMMUNITY

- **PhD Representative (2023–2024)**: Elected to represent >180 doctoral researchers at MPI CBS.
- **Mentoring**: Supervised working students and interns in the research group.
- **Talks**: Presented research findings at international conferences, internal institute colloquia, and interdisciplinary research meetings.